

CodeAlpha_DataVisualization

Project Title

Data Visualization using Titanic Dataset

Internship Domain

Data Analytics Internship by CodeAlpha

Objective

The objective of this project is to transform raw data into meaningful visualizations using Python libraries such as Matplotlib and Seaborn. The project helps in understanding trends, patterns, and insights from the Titanic dataset.

Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- VS Code

Dataset Used

Titanic Dataset

Features

- Data Cleaning
- Graphical Representation
- Statistical Visualization
- Trend Analysis
- Heatmap Visualization

Visualizations Included

- Survival Count Plot
- Gender vs Survival Chart
- Age Distribution Histogram
- Passenger Class Count Plot
- Correlation Heatmap

Key Insights

- Female passengers had higher survival rates.
- Passenger class influenced survival chances.
- Most passengers were between 20–40 years old.
- Fare and passenger class showed correlation patterns.

How to Run the Project

Install Required Libraries

```
pip install pandas matplotlib seaborn numpy
```

Run Python File

```
python data_visualization.py
```

Project Files

- data_visualization.py
- Titanic-Dataset.csv
- README.md
- screenshots/

Author

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